Six applications of one common theory: the lengths of the manuscripts

Human Compression project

11 October 2026

The specified main results of six historical manuscripts follow from one common theory and six complete applications. Their mathematical contracts have been independently reviewed and admitted in the published edition `6641c356`. The numerical audit is separate from the mathematical admission. All figures below bind that edition, including its final attribution correction.

Let c count NFC characters other than whitespace in the bound PDF extractions, after the reviewed page-furniture exclusions and glyph calibration. The comparison of written papers is

$$R_A = \frac{\sum_{i=1}^{6} c(L_i)}{c(U_A) + \sum_{i=1}^{6} c(A_i)} = \frac{195{,}379}{49{,}059 + 78{,}251} = 1.534671\ldots .$$

The common theory is paid in full once within this topic. Overlap with another topic does not reduce this charge. Including the entire required program, printed as one appendix P, gives the same overall comparison with that cost included:

$$R_{A,P} = \frac{\sum_{i=1}^{6} c(L_i)}{c(U_A) + \sum_{i=1}^{6} c(A_i) + c(P)} = \frac{195{,}379}{127{,}310 + 53{,}035} = 1.083362\ldots .$$

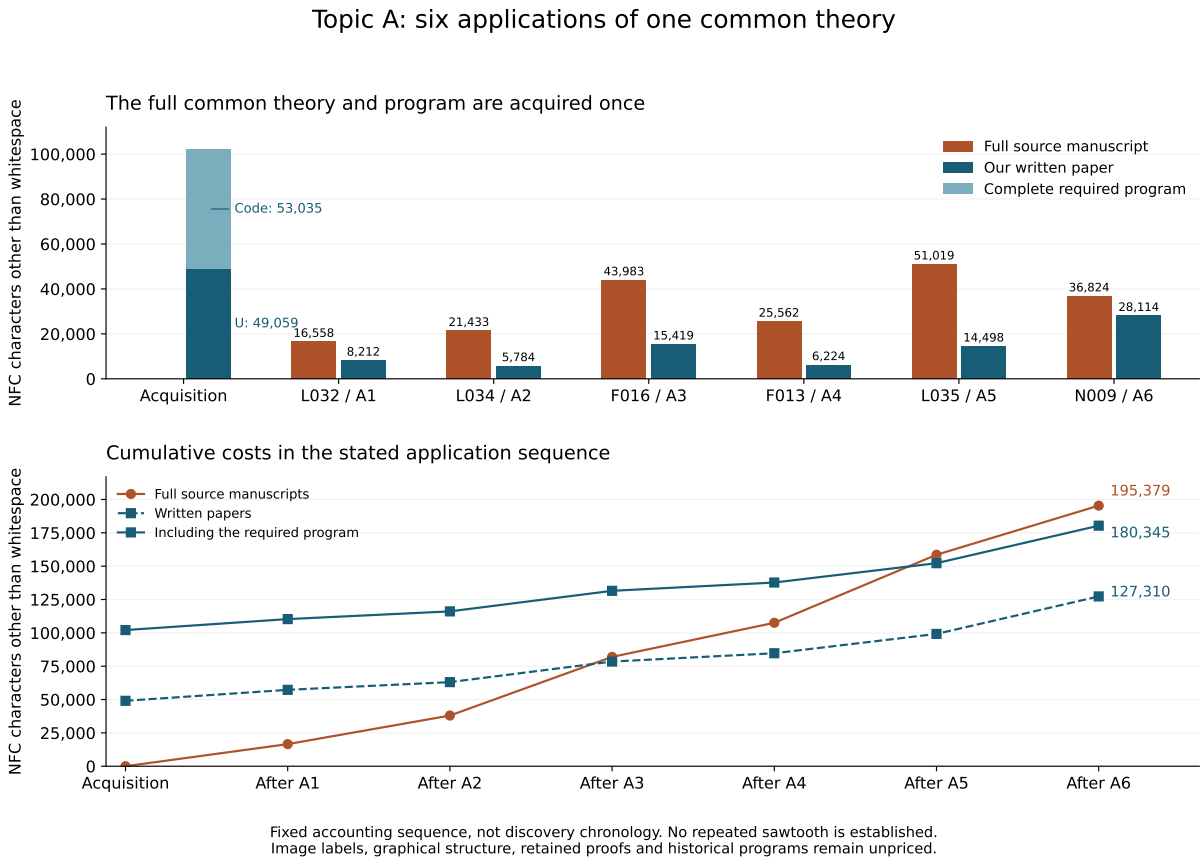
The four raw program files are not charged a second time.

Source	Complete application	Source	Application	Conditional
L032	A_1: two-power orders	16,558	8,212	2.016318
L034	A_2: three-generator hypertopes	21,433	5,784	3.705567
F016	A_3: square sections and extensions	43,983	15,419	2.852520
F013	A_4: common two-group quotient	25,562	6,224	4.107005
L035	A_5: high-type two-groups	51,019	14,498	3.519037
N009	A_6: order-1024 markings	36,824	28,114	1.309810
	All six applications	195,379	78,251	—
	Acquisition: whole common theory	—	49,059	—
	Acquisition: complete program	—	53,035	—

Each conditional factor is $c(L_i)/c(A_i)$ *after acquiring the whole common theory and the whole complete program appendix*. It does not measure the cost of proving that application independently. The overall comparison charges the common theory explicitly, and its version including the program also charges that complete appendix.

What is recovered and what is counted. The fixed contracts cover the all-parameter two-power existence theorem (L032), the specified three-generator hypertope presentations and geometry (L034), square sections, flatness, amalgamation and extension results (F016), the common lower-central quotient and covering theorem (F013), the high-type existence and complete marked classifications (L035), and the nondegenerate order-1024 marked classification and all its comparison tables (N009). Each contract preserves its reviewed literal corrections and retained inputs. The particular inductive presentation and separate classification question in L032, Question 3.3 in L034, and other statements outside the declared contracts are not asserted to follow from this volume. Their source text is nevertheless included in the full-manuscript numerator.

Every source page, title, abstract and bibliography is included. The denominator contains the entire common paper, all six complete applications and, for the second measure, the whole program appendix. There are 93 source pages, 17 common-paper pages, 33 application pages and 21 program pages. These physical extents are reported separately from the text coordinate.



The plotted sequence acquires the full common theory and program before the applications. It is an accounting order, not a chronology of discovery or evidence of a repeated sawtooth. Historical background has not been assigned a zero cost by the plotted starting value.

Metric sensitivity. No conversion between words, characters and bytes is presumed. Each row uses its own unit consistently on both sides, with whole-document costs summed without invented separators. The primary row is NFC characters other than whitespace.

Form	Unit	Written papers	Including program
NFC	Whitespace-separated tokens	1.770375	1.556223
NFC	Alphanumeric units	1.896848	1.444332
NFC	Characters other than whitespace	1.534671	1.083362
NFC	Characters with collapsed whitespace	1.577151	1.154279
NFC	UTF-8 bytes with collapsed whitespace	1.488189	1.122995
NFKC	Whitespace-separated tokens	1.772414	1.558015
NFKC	Alphanumeric units	1.896848	1.444332
NFKC	Characters other than whitespace	1.535608	1.084214
NFKC	Characters with collapsed whitespace	1.578266	1.155246
NFKC	UTF-8 bytes with collapsed whitespace	1.638395	1.206272

The machine-readable record additionally retains every untrimmed and raw selected coordinate. With no furniture removed, the primary factors are 197,304/127,370 = 1.549062 ... for written papers and 197,304/180,438 = 1.093473 ... including the program. Font-aware calibration does not change the primary nonspace count here, but it does change some alphanumeric and byte counts. The printed discretionary hyphens are retained throughout.

As an alternative representation of the *same* complete required program, the four raw files contain 51,970 primary characters. Replacing the printed appendix by those files gives 195,379/179,280 = 1.089798 .... The two representations are never added together.

Limits of the measurement. This is a comparison of extracted text, not complete visible text. L035 pages 9, 10, 12 and 22 contain raster scientific diagrams and unextracted labels; page 22 has no extracted text, and its embedded image is clipped. N009 page 1 contains an image-only journal banner. Their missing labels and graphical structure have unknown cost, not zero cost. No private image transcription is included in these figures.

Classical group theory, geometric correspondence and the stated source inputs remain retained. In particular, the exact infinite universal decomposition and the specified smooth-square classification used for N009 are not newly proved here. The full costs of retained proofs, historical external programs, libraries and graphical constructions remain unknown. The factors above therefore do not claim to price every artifact needed on both sides. The separate formal scope identifies which generic lemmas Lean verifies; neither an end-to-end formalization nor its cost is claimed.

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